

# Pramish Pandey

[pramish.pandey@usm.edu](mailto:pramish.pandey@usm.edu) — [pramishpandey.com.np](https://pramishpandey.com.np) — [linkedin.com/in/pramishpy](https://linkedin.com/in/pramishpy) — [github.com/pramishpy](https://github.com/pramishpy)

## Education

### The University of Southern Mississippi

Jan 2024 – Present

*Bachelor of Science in Computer Science*

- **GPA:** 4.00 / 4.00 — President's List (4 consecutive semesters)
- **Relevant Coursework:** Database Management Systems, Intro to AI, Machine Learning, Object-Oriented Programming, Algorithms, Computer Systems, Operating Systems, Secure Software Development, Computer Networks, Linear Algebra, Probability & Statistics, Data Governance Concepts.

## Experience

### Data Engineer & Full Stack Developer

September 2025 – Present

*School of Polymer Science & Engineering, The University of Southern Mississippi*

- Managing enterprise data models and ETL workflows using Python and SQL to efficiently query, clean, and synthesize more than 50,000 records to store in the custom designed database.
- Using Microsoft Excel and Python to compile, analyze, and present data analytics dashboards to researchers across polymer science.
- Designing a Django backend to organize and upload records efficiently in bulk to reduce manual data upload-time by 97%.
- Researching & Deploying the Django backend, React front-end and PostgreSQL database in GCP while documenting the entire workflow, tests, errors and presenting the overall progress in weekly meetings.
- Link to my work: <https://polymermorphology.org>

### AI4ALL Ignite Fellow (Applied AI & LangChain)

January 2026 – Present

*Center for Faculty Development, The University of Southern Mississippi*

- Selected for a highly competitive technology accelerator and dedicating 10+ hours weekly in researching Generative AI, Large Language Models (LLMs), and prompt engineering.
- Gained hands-on experience prototyping AI workflows utilizing Python and LangChain, and exploring enterprise integrations with cloud AI agents.

### Learning Assistant – College Algebra Support

August 2025 – December 2025

*Center for Faculty Development, The University of Southern Mississippi*

- Utilized strong communication skills to explain complex mathematical concepts and algebraic logic to diverse audiences, helping students troubleshoot problem sets effectively.
- Demonstrated acute attention to detail by reviewing student work for calculation errors and adherence to foundational algebraic principles.

## Projects

### CollabNet: Connecting Developers and Researchers

[Website](#)

- Architected a scalable front-end user interface utilizing Next.js and React, achieving a 98/100 Lighthouse performance score through optimized rendering.
- Designed relational data models and executed complex SQL queries to manage over 1,000+ concurrent data transactions using PostgreSQL.
- Implemented strict Row Level Security (RLS) policies to securely isolate 100% of user data to align with modern PII, PHI, and data governance standards.

### Mentally: Automated Infrastructure & CI/CD

[Website](#)

- Built an autonomous reporting and deployment infrastructure using GitHub Actions and Docker, implementing test automation to run unit tests and integration checks on every commit.
- Developed a Flask (Python) backend to handle API requests and data validation, ensuring 99.9% uptime and reliable service delivery for users.
- Integrated augmented reality (AR) components with backend services, requiring troubleshooting of complex interactions between browser hardware access and server-side logic.

## Skills

**Languages:** Python (Strong proficiency), TypeScript, C++, SQL, JavaScript.

**Tools:** LangChain, Generative AI, Git, PostMan, Docker, Kubernetes, AWS, GCP, Supabase, Neon, Claude Code, Visual Studio, PostgreSQL, Microsoft AI Studio(Azure), Microsoft Office Suite(Excel, Word, Project, PowerPoint).

**Core Competencies:** Data Governance (PII/PHI), Object-Oriented Design (OOD), Cross-Functional Collaboration, Data Structures & Algorithms, Data Visualization, System Architecture.

**Frameworks & Libraries:** React, Next.js, Django, Node.js, pandas, NumPy, OpenCV.